

Documentation Audit Report

Date: 2026-03-16

Branch: docs/audit-consistency-fixes

PR: #3 (<https://github.com/cnsp/ems-optimization/pull/3>)

Audit Scope

Comprehensive review of all 60+ documentation files (.md, .yaml) in the EMS Optimization repository to identify and fix inconsistencies caused by:

1. **P0 baseline migration** — The P0 policy changed from index-based uniform allocation (~8.1 min mean RT) to spatially-stratified allocation (~3.17 min mean RT), but many documents retained the old metrics.
 2. **Capacity default change** — Default firehouse capacity changed from 5 to 2 (per DEC-010), but several docs still referenced the old value.
 3. **Nomenclature standardisation** — “P0-spatial” interim name should be just “P0” (per DEC-012), but appeared in many analysis documents.
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Issues Found and Fixed

Category 1: Stale P0 Metrics (Critical)

The most pervasive issue. When P0 was redefined from index-based uniform to spatially-stratified, some document sections were updated while others retained old values, creating internal contradictions.

Document	Issue	Fix
executive_summary.md	68% RT reduction, 8.1 min P0, 64.2% coverage, P2 named “Maximal Coverage”, d=28.9	Rewrote with current P0 metrics (3.17 min, 99.7%, 19% improvement)
executive_presentation.md	Slide 9: 5.5 min delta, 167,750 min annual savings, 3× fleet multiplier	Corrected to 0.60 min delta, ~18,300 min annual savings
technical_report.md §5.2	ANOVA $F=12,010$, $\eta^2=0.996$	Corrected to $F=1,019$, $\eta^2=0.959$
technical_report.md §5.6	$\Delta=-5.51$ min, d=28.9	Corrected to $\Delta=-0.60$ min, d=10.3
technical_report.md §7.2	Finding 1 cited $F=12,010$	Corrected to $F=1,019$
technical_report.md §7.5	Impact table: d=28.9	Corrected to d=10.3
re-search_questions_assessment.md RQ3	-68.2% RT, d=28.9, $F=12,010$	Corrected to -18.9%, d=10.3, $F=1,019$
re-search_questions_assessment.md RQ5	P0 8-min coverage: 57.5%-99.5% (old uniform)	Regenerated from production_v2: 99.1%-99.8%
project_alignment_verification.md	“P2 reduces mean RT by 68.2% vs P0”	Corrected to 18.9%

Category 2: P0-Spatial → P0 Naming

Document	Occurrences Fixed
capacity_sensitivity_analysis.md	~25 occurrences
alternative_analyses_summary.md	1 occurrence
final_summary.md	3 occurrences
code_documentation.md	1 occurrence (+ clarified description)
executive_presentation.md	2 occurrences (Slides 3, 5)

Category 3: Capacity Default (5 → 2)

Document	Line	Fix
conceptual_model.md	§3.2	C=5 → C=2 (with sensitivity analysis note)
cbd_focused_optimization_analysis.md	Line 28	“Maximum 5 units” → “Maximum 2 units”
project_alignment_verification.md	Line 191	capacity ≤ 5 → ≤ 2
technical_report.md	§12.3 config table	capacity (5) → (2)

Category 4: Other Fixes

Document	Fix
executive_summary.md	Simulation count 1,770 → 2,700+
executive_presentation.md	Simulation count 1,440 → 2,700+
research_questions_assessment.md	Simulation scale updated; effect size reference corrected
policy_tradeoff_analysis.md	Added prominent deprecation/context note explaining this uses the old uniform P0
implementation_roadmap.md	Baseline label “P0, cap=5” → “P0, cap=2”
capacity_sensitivity_heatmap.png	Regenerated with 3 panels (K=20, 30, 40) instead of 2
generate_capacity_sensitivity_heatmap.py	k_values=(20, 40) → (20, 30, 40)

Documents Reviewed but Not Modified

The following documents were reviewed and found to be accurate in context:

- `nomenclature_migration.md` — Historical reference, correctly describes the transition
- `decisions_log.md` — Historical record, mentions old metrics in correct historical context
- `assumptions_log.md` — Already updated with A14/A15 for P0 nomenclature
- `experimental_design.md` — Correctly describes v1 experiments with cap=5
- `firehouse_capacity_analysis.md` — Analysis document, cap=5 references are to the analysis subject

- `queue_analysis.md` — Uses 1,440 run count correctly (production-only scope)
- `cbd_comparison_and_validity_report.md` — Historical record of CBD addition
- `project_workflow_wbs.md` — Work breakdown with historical task descriptions
- All config files — Already correct (optimization.yaml has firehouse_capacity: 2)

Verification

All corrected values were cross-referenced against:

- `results/production_v2/tables/descriptive_statistics.csv` (primary source of truth)
- `results/production_v2/tables/anova_results.csv`
- `results/production_v2/tables/effect_sizes.csv`
- `configs/optimization.yaml` (firehouse_capacity: 2)

Remaining Known Issues (Low Priority)

1. Some internal/historical docs (`gap_closure_report`, `cbd_comparison_report`, `project_workflow_wbs`) still reference “1,770 runs” or “1,440 runs” — these are accurate for their specific scope and historical context.
2. The `research_questions_assessment.md` Experiment 1 header says “cap=5” — technically correct as the experiment was designed with cap=5, but results are identical to cap=2 since no firehouse gets >2 units at K=20.
3. `technical_report.md` §4 experiment table lists “cap=5” for Exp1-4 — historically accurate for the original experiment design.